Foreign Direct Investment and Economic Growth in Chinese Special Economic Zones

Panel Evidence from 1996-2017

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October 2024

Both foreign direct investment (FD) and special economic zones (SEZs) lie at the heart of the Chinese economic growth project. Research on this topic has largely debated the bidirectional causal relationship between FDI inflows and growth or described the mixed effects of FDI on economic and social development, both at the national level. However the link between inbound FDI and SEZ economic growth itself has bee ignored. Using an original panel dataset covering 19 Chinese SEZs (and SEZ-like zones) over 1996-2017, we explore the link between FDI and regional economic growth using a series of econometric models. We also examine the impact of the 2008 financial crisis on this relationship. Our results indicate a significant positive effect of inbound FDI on economic growth in SEZs. Following the 2008 crisis, the effect of FDI inflows on economic growth was reduced, pointing to the possibility of a lasting reduction in the relationship between the two as a result of the crisis. Finally, we identify the implications of our results for promoting sustainable economic development in SEZs, as well as potential points for reform in Chinese FDI policies more broadly.

Conflicts of interest: none

Data access was supported by funding from the UCL School of Public Policy's departmental research fund.

1. Introduction

Since the beginning of Deng Xiaoping's Open Door Policy, special economic zones (SEZs) and foreign direct investment (FDI) attraction have formed core components of China's modern economic growth. The combination of the two in China has been remarkably successful: SEZs have been associated with a 58% increase in FDI per capita while not displacing domestic investment, additionally bringing about an increase in total factor productivity (Wang 2013). While research has examined the effects of both FDI and SEZs on national-level Chinese statistics with particular focus on spill-over effects (such as Zhang 2001; Chakraborty et al. 2017), the link between FDI and economic growth within the SEZs themselves has been ignored.

We address this gap in the literature, examining the relationship between FDI and growth in Chinese SEZs with an eye toward assisting policy makers in reaching their goals of pursuing sustainable economic growth through SEZ development. We analyse FDI's effects on growth over a two-decade period across nineteen SEZs and SEZ-like zones, following previous practice (such as Graham 2004; McCallum 2011). In addition to addressing the relationship between inward FDI and economic growth within SEZs, we explore the impact of the 2008 financial crisis and Great Recession on this link. We find that FDI has positively contributed to economic growth in Chinese SEZs, in contrast with the SEZ experience outside China. The 2008 crisis itself appears to have had no significant effect on the relationship between FDI and GDP.

2. Special Economic Zones and FDI

An SEZ can be described as a delimited geographical region that is managed and administered in a manner that differs from those in other parts of the same country, usually involving a duty-free environment and investment-friendly regulations and policies. The rationale behind SEZs is that their business-friendly environment can provide a stimulus for national economic development.

Historically, SEZs established prior to the 1970s across the world were generally intended to accommodate developing countries' import-substitution policies (Farole and Akinci 2011). However, with the disappearance of import-substitution industrialization, these zones have largely been converted into the core policy instruments for trade and FDI in several developing countries—especially East Asia and Latin America—in order to attract multinational companies to invest in their labour-intensive manufacturing industries. In this post-debt crisis context, an abundance of empirical work has demonstrated SEZs' positive influence on development through heightened employment rates (Farole and Akinci 2011; Jayanthakumaran 2003; Mongé-Gonzalez et al. 2005).

However, more recent effects of SEZs have proven to be heterogeneous, reflecting a wider range of policy goals and economic environments. In the case of China, SEZs went hand in hand with investment attraction in support of export-led growth through export-oriented manufacturing, with the three forming something of a positive-feedback loop (Ali and Guo 2005; Long 2005). However, local conditions meant that policies that had seen success in one SEZ would not necessarily translate into success in another (Chou and Ding 2015), although Crane et al. (2018) discuss ways to leverage SEZs' policy-driven successes elsewhere in China. Moreover, outcomes across countries are even more variable. For example, India's SEZs have not been associated with developmental spillovers (Alkon 2018), while the effect of FDI on economic growth across the economy varies by sector (Chakraborty and Nunnenkamp 2008). These sorts of results have contributed to a wariness to recommend SEZs as a development strategy (FIAS 2008; Farole and Akinci 2011; Zeng 2012, 2016).

2.1 The Development of Chinese SEZs

The historical development of Chinese SEZs and SEZ-like zones has its roots in the 1970s, as a response to the failure of protectionist and inward-focused foreign economic policies to stimulate economic growth (Ying and Trautwein 2013). A combination of liberalizing reforms to domestic and foreign economic policy stimulated FDI inflows focused particularly in light industry (Ali and Guo 2005; Davies 2012). The SEZ program was introduced in 1979 to further focus inbound investment and growth in relatively poor and underdeveloped coastal regions with local industrial potential (Leong 2012; Crane et al. 2018; Crane 2019). The success of these new zones cemented the political power of reformers in the Central Committee, enabling the establishment of further SEZs (Moberg 2015, 2017). Following the Tiananmen Square incidents FDI inflows fell, although growth returned by 1991. Subsequently, new reforms aimed at attracting further FDI were promulgated, while additional SEZs along China's southern coast were established, bringing the total to nineteen. Graham (2004) has described the geographic preference for coastal regions as attractive for investors due to higher quality infrastructure than that found in other regions, as well as easy access to airport and seaport facilities.

Even as China joined the World Trade Organization (WTO) in 2001 and remained an attractive destination for FDI, new incentives were introduced to increase the attraction of its SEZs. However, these incentives were met with a backlash from foreign enterprises and operations located outside the SEZs, with complaints of discrimination, leading to an end of national business-tax exemptions for foreign-funded firms in 2007. In place of these exemptions, a new series of incentives was implemented, involving fiscal inducements, five-year tax holidays, and a

50% reduction in business income tax for foreign investors (Yuan 2017; Crane 2019; Gao and Guo 2019). These policies went hand-in-hand with a broadly favourable environment to private enterprise. As Moberg (2017) finds, the benefits are wide ranging, as SEZs are associated with increased employment, technology transfers, and improved labour practices, enterprise management technology, and infrastructure.

2.2 Linking FDI Inflows to Economic Growth in Chinese SEZs

The narrative of the general success of Chinese SEZs has been built upon their economic effects for the broader Chinese economy. SEZs have increased capital inflows and exports, and have contributed to national economic growth. However, these effects are often captured as spillovers to neighbouring regions or as contributions to national accounts. A small number of studies do examine questions relating to FDI and economic growth at the regional level, albeit without particular attention to the policy environment encapsulated in the SEZ, although these are somewhat dated. Sun and Parikh (2001) focus their analysis primarily on exports and growth, finding the relationship to be substantially mediated by the market-oriented policies. Yao (2006) finds FDI and exports positively impact economic growth across Chinese regions. Some studies have focused on specific SEZs, such as Chung's (1999) examination of Qingdao and Chou and Ding's (2015) attempt to draw lessons from Shenzhen's successes for Kashgar.

When examining FDI's influence on economic growth in Chinese SEZs, we can draw insights from a vibrant body of research investigating FDI and growth in the developing world. Investment has long figured prominently in economic growth models. Borensztein et al. (1998) posit that FDI's contribution to economic growth is a result of technology transfers, the success of which is determined by the host country's absorptive capacity. Turning to China, indirect evidence of investment spillovers in Chinese SEZs, with evidence of knowledge- and technology-based advantages over non-SEZ regions (Moberg 2017). Within SEZs, this is likely to be encouraged through policies favouring market liberalization. As a result, we can expect FDI to positively contribute to economic growth in Chinese SEZs.

H_1: FDI inflows positively influence SEZs' GDP.

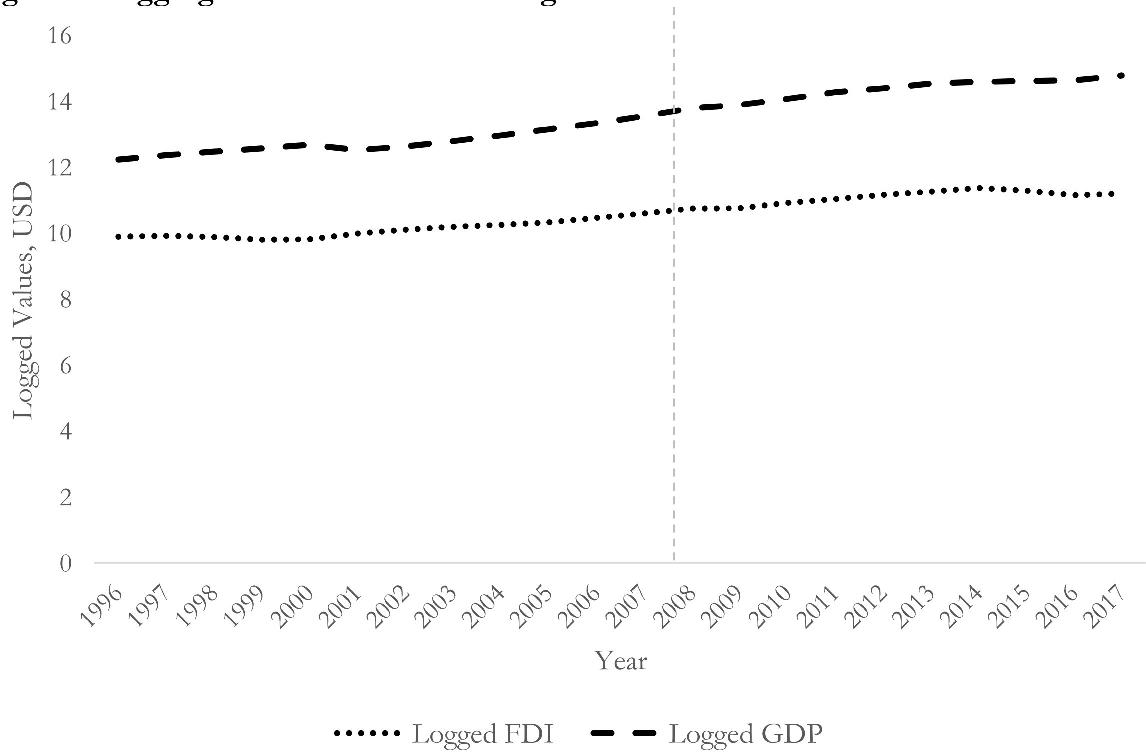
The export-oriented nature of China's SEZs makes them particularly vulnerable to global shocks, which may affect the relationship between FDI and economic output. While the US-China trade war and the COVID-19 pandemic fall outside the temporal scope of our sample, we are able to capture the 2008 financial crisis. Particularly notable here was the significantly reduced final-product consumption in developed economies, which contributed to what has been referred to as the Great Trade Collapse, as supply chain linkages multiplied the effects of reduced developed-economy imports (Baldwin 2009; Bown 2018). At the same time, credit tightening in many originating countries for FDI reduced FDI outflows to markets like China (Poulsen and Hufbauer 2013).

While the Chinese economy itself continued to grow at a steady pace, the external orientation of SEZ economies placed them at risk of slowdown as a result of their economic ties with developed economies, potentially disincentivising future FDI. Additionally, contractionary pressures on the supply of FDI would likely reduce its agglomeration effects, inhibiting the economies of scale and spillovers that promote its broader effect on regional economic growth. Consequently, post-crisis growth would likely be driven by domestic inputs to a greater extent than international sources when compared to the pre-crisis period.

H_2: External contractionary pressures during the financial crisis reduced FDI's effects on SEZs' GDP.

Figure 1, below, presents temporal trends in the cumulative figures for FDI and GDP in our sample. Throughout the period, the two measures appear to track each other quite closely. FDI inflows stagnate between 2008 and 2009, increasing only 0.6% after several years of very strong growth. GDP, on the other hand, experienced no slowdown, resulting in FDI inflows becoming an ever smaller portion of GDP.

Figure 1 – Aggregate FDI and GDP among Chinese SEZs



3. Data and Analytical Methods

Our data cover a panel of nineteen SEZs (the complete list is available in the appendix), inclusive of the SEZ-like zones established in the 1980s, over the period 1996-2017. The sample's starting point was chosen due to data availability for several of the key series, while the 2017 endpoint avoids the profoundly disruptive effects of the US-China trade war the onset of the COVID-19 pandemic. While the consequences of these events on SEZ performance are worthy of investigation, they fall outside our focus on the relationship between FDI and GDP.

Data were sourced from the Bureau of Statistics' National Database, CEIC, and province- and municipality-level statistical yearbooks. While the national database is one of the most comprehensive and detailed official online sources of Chinese statistics, it does not record data for some small SEZs, such as Qinhuangdao, Lianyungang and Behai. The statistical yearbooks are used to augment the some of the time series available through the national database (depending on the series, national statistical coverage could range from 1999-2017 or 2007-2017), although sector-level value-added data were unavailable for Lianyungang and Qinhuangdao, leaving us with seventeen SEZs in our empirical models.

Table 1 presents the variables employed in our study, all of which are reported on an annual basis. The key outcome that we seek to explain is economic output, as captured by regional gross domestic product (GDP), while the primary explanatory variable of interest is FDI. We control for population size, sector-level value added, population, average urban income, fixed asset investment, trade, and public revenues. Support for these additional variables can be found in the literature: the indicators of economic activity (sectoral value added, investment, and trade) are all directly related to inputs in growth models, while population provides a measure of an SEZ's size, including its potential workforce. We include public revenues because of fiscal policy's active role in promoting regional economic growth.

Table 1: Variable Descriptions

	Construct	Variable name	Unit
Dependent Variable	Gross Domestic Production	GDP	Millions, USD
Independent Variable	Inward Foreign Direct Investment	FDI	Millions, USD
Control Variables	Value-Added of the Primary Industry	PI	Millions, USD
	Value-Added of the Secondary Industry	SI	Millions, USD
	Value-Added of the Service Industry	SER	Millions, USD
	Total Population	POP	Millions of people
	Average Income of Urban Workers	INC	USD
	Fixed Asset Investment	FI	Millions, USD
	Total Exports	EXPORTS	Millions, USD
	Public Finance Revenue	REV	Millions, USD

Figures originally reported in CNY were converted to USD using the annual official exchange rate.

China's rapid economic development has raised concerns among observers over the potential erosion of its low-cost labour advantage, which may place a damper on growth. Conversely, multinationals tend to be able to pay higher wages than domestic employers due to their superior productivity. Likewise, higher incomes will typically be associated with highly skilled employment in the secondary and tertiary industries that have served as engines for growth in China's SEZs. We include urban income in light of these contradictory expectations. To investigate H_2, we create a binary indicator, POSTCRISIS, to capture the post-crisis period, taking a value of one over the years 2009-2017, and zero for the preceding years in the sample.

Table 2 presents summary statistics. In our models, these variables are log-transformed to generate approximately normal distributions. Table A1, in the appendix, contains bivariate correlation coefficients. Altogether, the dataset is structured as a narrow panel or short time-series cross-section, structured as an unbalanced panel of 275 observations across 17 economic zones with a gap in Hainan's reporting (Lianyungang and Qinhuangdao do not report key variables and are omitted from the sample).

Table 2: Summary statistics, log-transformed variables

Variable (logged)	Abb.	N	Mean	St Dev	Minimum	Maximum
GDP	GDP	418	10.090	1.324	7.006	13.064
FDI	FDI	413	6.830	1.538	2.896	9.959
Primary VA	PI	301	7.139	1.255	4.324	9.960
Secondary VA	SI	301	9.310	1.582	5.669	11.875
Services VA	SER	301	8.866	2.381	0.001	12.696
Population	POP	412	15.390	0.718	13.390	16.493
Urban Income	INC	390	8.258	0.945	6.515	11.152
Fixed Investment	FI	418	9.041	2.172	0.745	16.009
Exports	EXPORT	393	13.116	2.581	3.701	17.236
Gov Revenue	REV	414	7.767	1.626	4.136	11.536

All variables are natural log transformed from their reported values. *Services VA* and *trade* had one added to their initial unlogged values so that their unlogged minimums were greater than one.

3.3 Empirical Approach

To examine the relationship between FDI and GDP (H_1), we employ a pair of linear models that incorporate unit-specific effects in different ways, enabling us to capture clustering by SEZ. The first is a typical unit (SEZ) fixed-effects model, yielding coefficients reporting within-unit effects.

Our second approach builds on the Mundlak model (Mundlak 1978; Allison 2009; Bell and Jones 2015; Schunck and Perales 2017), which allows for the estimation of both within-cluster and between-cluster effects. By distinguishing cluster means from within-cluster deviations from the cluster means, this approach enables a fuller understanding of the ways in which differences across SEZs shape the relationship between FDI inflows and economic growth. Demeaning at the cluster level allows for estimation of higher-level – between-cluster – effects by reducing dependency between the error term and cluster-specific effects. The resulting model takes this form:

$$y_{ij} = \alpha + \beta_1(x_{ij} - \bar{x}_i) + \beta_2\bar{x}_i + \mu_{ij} + \varepsilon_{ij} \qquad 1$$

As described above, β_1 captures the effect of demeaned within-cluster deviations from cluster means for independent variables that exhibit variation both within and between clusters. Between-cluster effects are captured through the coefficient of the cluster mean itself, β_2, while μ_{ij} represents random cluster-specific intercepts, α is a common intercept, and ε_{ij} is the error term. β_1, the coefficient for the within effect, is mathematically equivalent to its counterpart in a standard fixed-effects model. Evidence from simulations demonstrates equivalent performance between fixed-effects and within-between models in finite samples (Bell and Jones 2015; Dieleman and Templin 2016).

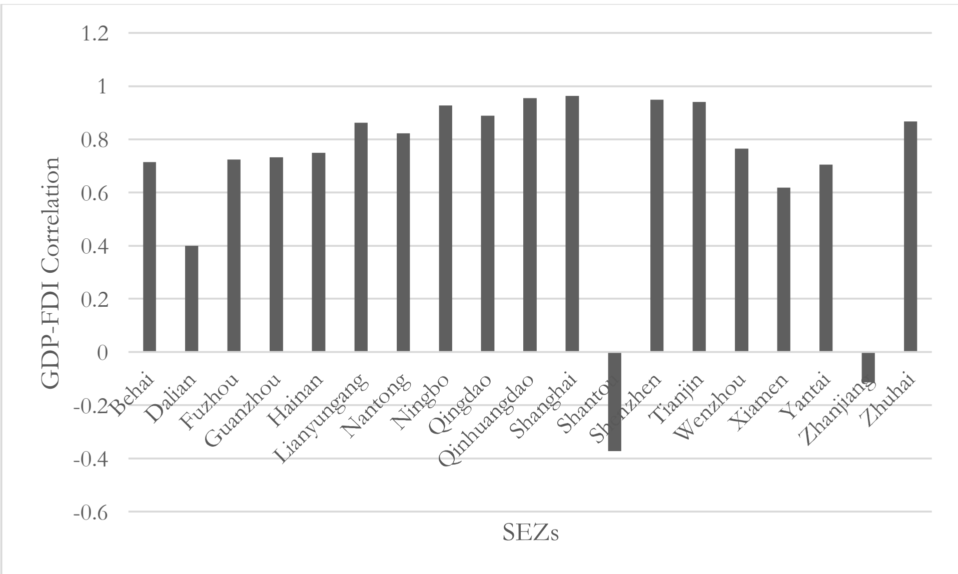
We take a similar approach to test whether the global financial crisis reduced the effects of FDI on GDP (H_2), adding the POSTCRISIS indicator to the model. We then interact this with FDI, which provides an estimate of the change in the effect of FDI on GDP in the post-crisis period of the sample. For this test, we rely on the fixed-effects model: the application and interpretation of interaction effects are established in this context, in contrast with the within-between model. We discuss an application to the within-between model and results in Section 4.3 and the appendix.

As both serial and cross-sectional correlation are present in the data, we employ the approach to estimating standard errors proposed by Driscoll and Kraay (1998), as these are robust to both spatial and serial correlation and are also consistent in the presence of heteroskedasticity. An additional benefit of the Driscoll-Kraay method is that, while its asymptotics rely on the panel's temporal dimension (T), simulation performance demonstrates strong performance even with short-T panels that exhibit cross-sectional dependence (Hoechle 2007).

4. Results

Cross-border economic integration is a central component to the success story of Chinese SEZs. However, this general impression masks heterogeneity across SEZ experiences. Figure 2 demonstrates this variation in bivariate correlations between FDI inflows and GDP across SEZs. While most SEZs present a strong positive correlation between the two, both Shantou and Zhanjiang exhibit a negative relationship between FDI and GDP; this stands in stark contrast to SEZs like Shanghai and Shenzhen, for which the correlation is greater than 0.95.

Figure 2: SEZ-Level Correlations between GDP and FDI



4.1 Examining the Relationship between FDI and SEZ GDP

This apparent relationship between FDI and GDP is reflected in our results in Table 3, which provide a test for H_1. Model 1, which incorporates SEZ fixed effects, demonstrates a positive link between FDI and GDP, with a 1% increase in FDI associated with a 0.02% increase in GDP. While both of our other measures of economic flows, FI and EXPORT exhibit positive coefficients, neither gains statistical significance.

When looking at the division of economic activity, PI is negatively associated with GDP, while both SI and SER exhibit positive links. The demographic indicators, POP and INC, both have positive coefficients, as does REV. With the exception of POP, all of these effects are statistically significant.

Table 3: FDI's Effects on SEZ GDP

	Model 1 (Fixed Effects)			Model 2 (Within-Between)	
	Coefficient	Standard Error		Coefficient	Standard Error
FDI	0.023**	0.011		*Within Effects*	
PI	-0.117***	0.044	FDI	0.023**	0.009
SI	0.316***	0.026	PI	-0.116***	0.042
SER	0.070**	0.032	SI	0.316***	0.027
POP	0.057	0.037	SER	0.067**	0.031
INC	0.438***	0.077	POP	0.057	0.039
FI	0.007	0.006	INC	0.443***	0.074
EXPORT	0.018	0.015	FI	0.007	0.006
REV	0.200***	0.047	EXPORT	0.018	0.016
			REV	0.199***	0.046
				Between Effects	
			FDI	0.257**	0.100
			PI	-0.034	0.112
			SI	0.207***	0.139
			SER	-0.056***	0.021
			POP	0.440*	0.267
			INC	0.272	0.609
			FI	-0.031	0.051
			EXPORT	0.029*	0.017
			REV	0.225	0.305
N	275			275	
TSS	160.77			171.84	
RSS	1.105			1.138	

*** p < 0.01, ** p < 0.05, * p < 0.10 Model 1 includes SEZ fixed effects, which are not reported. Standard errors are estimated using the Driscoll-Kraay method.

Model 2 presents the within-between results. The within coefficients represent intra-cluster effects; as expected, these are nearly identical to those in Model 1, apart from slight rounding differences. The between effects identify the drivers of variations in GDP outcomes across SEZs. The FDI effect is large and positive, with 1% more FDI associated with nearly 0.26% greater GDP. Following the SEZ-level correlations between FDI and GDP in Figure 2, this result is unsurprising and seems to point to the importance of agglomeration effects.

SI, POP, and EXPORT all exhibit positive and significant effects on GDP. In contrast, SER has a negative and significant effect, likely highlighting heterogeneity across SEZs when it comes to services-sector composition and value-added contributions. When viewed in comparison with the SI result, this points to the key role played by manufacturing in driving Chinese SEZs' growth.

4.2 Is There a Financial Crisis Effect?

Our second set of tests investigates the effects of the 2008 financial crisis on the FDI-GDP nexus, as described in H_2, presented in Table 4. When POSTCRISIS is added to the model, without any interaction, its coefficient is positive and significant, pointing to a near 8% increase in average logged GDP following the 2008 financial crisis. When the interaction with FDI is added in Model 4, POSTCRISIS's coefficient halves in magnitude and loses significance. The interaction itself is very small in magnitude and insignificant, indicating a lack of change in FDI's influence on GDP between the two periods.

In short, we find evidence that FDI contributes to GDP in China's SEZs (H_1), while, in contrast with our expectations, this relationship does not appear to have been significantly impacted by the effects of the 2008 financial crisis (H_2). The results among our other covariates reflect well-established characterizations of China's SEZs: an emphasis on manufacturing, cultivation of skilled urban workforces, and active growth-oriented fiscal policies. It is also worth noting evidence of heterogeneity across SEZs, as indicated in Model 2's between effects.

Table 4: FDI and SEZ GDP, Financial Crisis Effects

	Model 3 (FE)		Model 4 (FE)	
	Coefficient	Std Error	Coefficient	Std Error
Postcrisis	0.079**	0.037	0.047	0.042
FDIXPostcrisis			0.004	0.003
FDI	0.027**	0.010	0.024***	0.009
PI	-0.151***	0.041	-0.146***	0.042
SI	0.344***	0.023	0.345***	0.023
SER	0.057*	0.030	0.052	0.033
POP	0.050	0.033	0.051	0.034
INC	0.418***	0.065	0.418***	0.065
FI	0.004	0.005	0.004	0.005
EXPORT	0.024*	0.013	0.027**	0.013
REV	0.183***	0.045	0.183***	0.045
N	275		275	
TSS	160.77		160.77	
RSS	1.015		1.012	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ All models include SEZ fixed effects, which are not reported. Standard errors are estimated using the Driscoll-Kraay method.

We test an alternative approach to estimating standard errors. Looking beyond the Driscoll-Kraay method, Beck and Katz's (1995) panel corrected standard errors (PCSEs) provide alternative to handling cross-sectional correlation and unit heterogeneity through the application of a sandwich-type covariance estimator to an OLS model. We apply PCSEs to a pooled OLS model and one with unit fixed effects and the results, presented in Table A3 in the appendix, are consistent with those presented above in Table 2.

While two-way FE models are commonly employed in TSCS and panel contexts, as Kropko and Kubinec (2020) highlight, the intended quantity of interest – impact of the variable(s) of interest on the outcome having accounted for otherwise unobserved unit and temporal effects – is statistically undefined. The reported coefficients from such a model are consequently both complex to interpret and unrelated to the quantity of interest we seek to estimate.

There is also the question of endogeneity. The narrow structure of the panel inhibits an application of generalized method of moments, and we are unaware of viable external options for instrumentation. We leave the question of causality to future researchers.

5. Conclusions

A sizeable body of research investigates the spillover effects of FDI into Chinese SEZs on neighbouring regions. We provide a first look at the influence of FDI on output within these SEZs themselves. We find that FDI is positively correlated with GDP within SEZs, although the magnitude of this effect is dwarfed by some characteristics of these regions' structure of economic activity. FDI is also positively correlated with GDP between SEZs. However, we do not find evidence of an effect of the 2008 financial crisis on the FDI-GDP relationship.

Looking beyond FDI, we find consistent evidence of the reliance on manufacturing in driving economic activity within SEZs. This is unsurprising, as are the influence of fixed investment and government expenditures. Indeed, these correspond with the common narrative around the success of Chinese SEZs. However, the between effects hint at variations in the SEZ experience, implying that caution may be warranted when making broad generalizations about the effects of these initiatives.

This exploratory study highlights interesting pathways for future research. Beyond the question of causality, there appears to be a fair amount of heterogeneity in SEZs' experiences with harnessing FDI inflows for economic growth. Variations across SEZs have not received much attention from researchers, as region-focused studies have instead generally opted to compare SEZs to non-SEZ regions. When looking beyond the Chinese economy, the common tendency has been to describe the Chinese SEZ experience as homogeneously successful, either implicitly or explicitly. Case-focused policy analysis of the experiences of lesser known and less successful SEZs can be useful informing policy recommendations for SEZs across the world.

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Appendix

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Table A1: Correlation Coefficients

	GDP	FDI	PI	SI	SER	POP	INC	FI	EXPORT	REV
GDP	1.000									
FDI	0.813	1.000								
PI	0.288	0.134	1.000							
SI	0.853	0.685	0.480	1.000						
SER	0.341	0.292	0.712	0.440	1.000					
POP	0.498	0.348	0.409	0.353	0.391	1.000				
INC	0.881	0.621	0.259	0.743	0.334	0.180	1.000			
FI	0.811	0.688	0.318	0.693	0.371	0.533	0.692	1.000		
EXPORT	0.453	0.326	-0.005	0.370	0.008	-6×10^{-4}	0.484	0.387	1.000	
REV	0.953	0.713	0.237	0.794	0.334	0.433	0.903	0.762	0.452	1.000

Table A2: PCSE Models

	Model A1 (Fixed effects)		Model A2 (Random effects)		Model A3 (RE, Driscoll-Kraay)	
	Coefficient	Std Error	Coefficient	Std Error	Coefficient	Std Error
FDI	0.023***	0.009	0.044***	0.010	0.044***	0.009
PI	-0.117***	0.031	-0.073***	0.027	-0.073***	0.028
SI	0.316***	0.030	0.309***	0.023	0.309***	0.013
SER	0.070**	0.030	-0.003	0.015	-0.003	0.014
POP	0.057**	0.027	0.171***	0.028	0.171*	0.099
INC	0.438***	0.037	0.465***	0.033	0.465***	0.077
FI	0.007	0.006	0.013**	0.006	0.013*	0.007
EXPORT	0.018	0.011	0.014	0.010	0.014**	0.006
REV	0.200***	0.017	0.206***	0.019	0.206***	0.048
N	275		275		275	
Adjusted R^2	0.992		0.991		0.991	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ Standard errors are based on the observed information matrix for Models A1 and A2. Model A3 employs Driscoll-Kraay SEs on random effects for comparison.

Table A3: PCSE Models

	Model A3 (No unit effects)		Model A4 (Unit fixed effects)	
	Coefficient	Std Error	Coefficient	Std Error
FDI	0.180***	0.010	0.028***	0.009
PI	-0.026*	0.015	-0.117***	0.038
SI	0.184***	0.014	0.316***	0.030
SER	-0.048***	0.006	0.070**	0.030
POP	0.437***	0.024	0.057**	0.027
INC	0.610***	0.038	0.438***	0.037
FI	-0.006	0.008	0.007	0.006
EXPORT	0.014***	0.004	0.018	0.011
REV	0.165***	0.024	0.200***	0.017
N	275		275	
Adjusted R^2	0.984		0.998	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ Both models are estimated using OLS and PCSEs. Model A2 incorporates a vector of fixed effects for SEZs, which are not reported.

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